CAR Planner: Constrained-Attention-Based Robust Imitation Learning for Autonomous Driving

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***Abstract*—In Transformer-based imitation-learning planners, ego-state attention can become excessively concentrated on a small subset of input channels. We propose CAR Planner, which imposes an inequality constraint on ego-state attention dispersion. A positive margin permits task-dependent, non-uniform attention while limiting excessive concentration. An augmented Lagrangian formulation adaptively adjusts constraint pressure during training without additional inference-time computation. On nuPlan, CAR Planner reduces attention concentration and improves closed-loop performance in several challenging settings relative to the evaluated learning-based baselines. Controlled multi-seed comparisons further show higher mean planning scores than fixed-coefficient attention regularizers, supporting adaptive attention constraints as a lightweight training approach.**

***Index Terms*—Motion and Path Planning, Imitation Learning, Planning under Uncertainty, Deep Learning Methods**

I. INTRODUCTION

ROBUST motion planning remains a central challenge in autonomous driving [1], [2]. Among data-driven approaches, imitation learning (IL) learns policies directly from expert demonstrations and has shown strong performance in modern driving benchmarks [3]–[5]. However, studies on shortcut learning highlight that reliance on spurious correlations can undermine robustness [6]–[8]. This motivates examining how IL planners allocate and use input information in challenging closed-loop scenarios.

In IL-based Transformer planners on nuPlan [9], ego-state attention can become concentrated on a small subset of channels. In this work, the current ego state comprises six channels, (x, y, ψ, v, a, s), representing position, heading, velocity, acceleration, and steering angle. Such concentration may reflect task-dependent selectivity, but excessive concentration can coincide with vulnerability when a dominant input channel becomes unreliable. The challenge is to limit excessive concentration while retaining flexible channel allocation. Existing approaches include fixed-coefficient penalties and the State Dropout Encoder (SDE) [10]. Fixed penalties apply a preset regularization scale, while SDE stochastically masks input-state channels during training.

We propose **CAR Planner**, which retains the input channels and adaptively regulates excessive ego-state attention concentration through constrained optimization. An inequality constraint bounds the deviation of attention weights from a uniform distribution, with a positive margin permitting task-dependent selectivity. The Augmented Lagrangian Method (ALM) [11], [12] adaptively adjusts constraint pressure according to the attention-dispersion constraint during training, while preserving the inference architecture. Evaluation on nuPlan combines attention analysis and controlled regularizer comparisons with open-loop score (OLS), non-reactive closed-loop score (NR-CLS), and reactive closed-loop score (R-CLS).

The main contributions of this work are as follows.

- **Constrained ego-state attention.** We formulate ego-state attention dispersion as an inequality-constrained optimization problem and enforce it with ALM to limit excessive single-channel routing.
- **Lightweight, architecture-preserving regularization.** The proposed constraint is applied only to the ego-state aggregation module and requires no auxiliary data, while preserving inference-time computation.
- **Benchmark and controlled evaluation.** CAR Planner combines competitive nuPlan benchmark performance with higher mean scores than fixed-coefficient attention regularizers in controlled multi-seed comparisons.

II. RELATED WORK

Autonomous driving planners are commonly grouped into rule-based, learning-based, and hybrid approaches. Rule-based methods such as IDM [13] and PDM-Closed [14] incorporate explicit driving rules and structured trajectory evaluation. Learning-based planners learn policies directly from data, with representative models including RasterModel, UrbanDriver [15], GC-PGP [16], and PDM-Open [14]. Hybrid planners such as GameFormer [17], PDM-Hybrid [14], and Pluto [18] combine learned predictions with structured planning components to account for interactions and trajectory feasibility. Other approaches incorporate auxiliary objectives, reinforcement learning, adversarial training, or closed-loop training [8], [19]–[22] to improve policy learning and address distribution shift.

The use of vehicle-state information requires attention to how the policy depends on these inputs. Causal confusion can impair imitation-learning policies under distribution shift [23]. State-token propagation, auxiliary learning, and state augmentation have been used to address inertia and offline–online performance discrepancies [24]. These studies motivate examining channel dependence alongside planning performance.

Recent IL planners increasingly use Transformer architectures [25] to integrate ego-state, surrounding-agent, and map features. Within nuPlan-based planning, PlanTF [10] applies the State Dropout Encoder (SDE) to stochastically mask input-state channels during training. CAR Planner instead retains the input channels and constrains excessive concentration in ego-state attention while preserving the inference architecture.

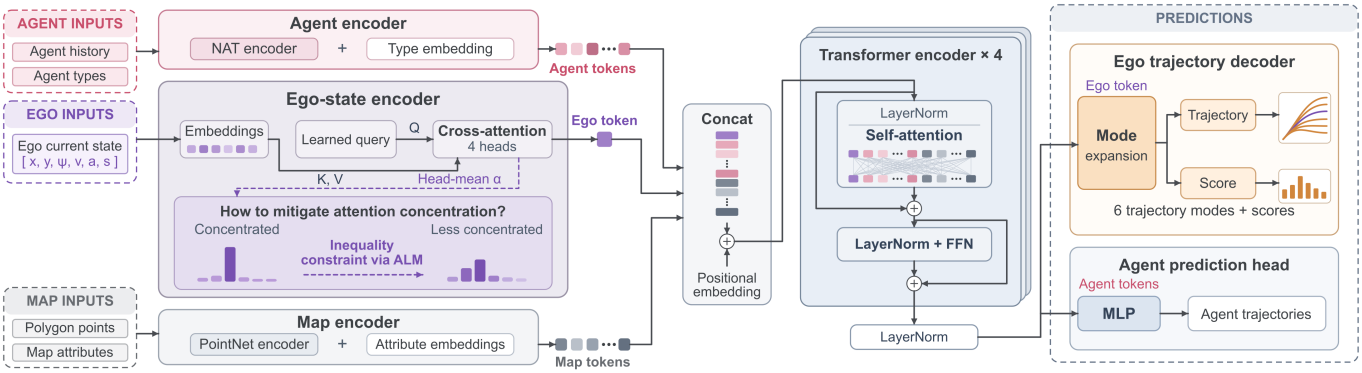


Fig. 1. Overview of the proposed **CAR Planner** architecture. Agent history, the current ego state, and local map polygons are encoded into agent, ego, and map tokens. The ego-state encoder uses a learned latent query with an attention constraint to regulate channel aggregation, and the resulting tokens are processed by a shared Transformer encoder for ego and non-ego trajectory prediction.

Attention regularization has also been studied for representation diversity and training stability. Disagreement regularization encourages diversity among attention heads [26]. Reparameterization has also been used to mitigate attention entropy collapse [27]. Constrained learning offers a complementary approach. Non-convex constrained learning has been formulated through a proxy-Lagrangian game [28]. CAR Planner applies projected ALM to an inequality constraint on ego-state channel concentration. Unlike fixed-coefficient attention penalties, its constraint pressure adapts during training.

Attention allocation and output influence are distinct aspects of model behavior. In NLP tasks, discrepancies have been observed between attention weights and gradient-based importance [29]. Attention perturbation analyses examine prediction changes induced by manipulating attention weights [30]. These studies motivate complementary analyses when interpreting attention distributions. CAR Planner uses attention as a lightweight training variable and examines its relationship to channel influence through contribution, Jacobian, and input-intervention analyses.

III. CAR PLANNER

A. Planner Architecture

Figure 1 illustrates CAR Planner's token-based architecture, consisting of input encoders, a shared Transformer, and trajectory prediction heads.

Common backbone. Agent-history tokens are produced from agent kinematic sequences, the ego token from the current 6-dimensional state (x, y, ψ, v, a, s), and the map from polygon features encoding lane geometry and semantics. The ego-state encoder embeds each scalar channel with a learned channel encoding. A learned query aggregates these embeddings through four-head cross-attention, whose head-averaged weights define the constrained distribution. For each token, a small MLP projects the pose descriptor $[x, y, \cos\theta, \sin\theta]$ into a geometric embedding. The resulting tokens are concatenated and processed by a shared Transformer encoder with key-padding masks for invalid inputs.

Prediction heads. From the encoded ego token, a trajectory head predicts M candidate future trajectories, each with T states $(x, y, \cos\psi, \sin\psi)$ and a trajectory-level mode score. At inference time, the most probable mode is selected and the heading is recovered by $\operatorname{atan2}(\sin\psi, \cos\psi)$. In parallel, a lightweight MLP predicts future positions for non-ego agents.

Difference between variants. The three planners differ only in ego-state regularization before the shared Transformer. **Base** uses no additional regularization, **PlanTF** applies State Dropout Encoder (SDE), and CAR Planner imposes a dispersion constraint on the same attention distribution. All other components are shared.

Default configuration. Unless otherwise specified, the shared backbone uses d=128, L_e=4 Transformer layers, h=8 Transformer self-attention heads, M=6 trajectory modes, and T=80 future steps.

B. Objectives

The following objectives are shared by Base, PlanTF (SDE), and CAR Planner. Let $\widehat{Y} \in \mathbb{R}^{M\times T\times 4}$ and $\pi \in \mathbb{R}^{M}$ denote the predicted ego trajectories and mode logits, respectively, and let $Y \in \mathbb{R}^{T\times 4}$ be the ground-truth ego trajectory. For non-ego agents, let $\widehat{P} \in \mathbb{R}^{(A-1)\times T\times 2}$ and $P \in \mathbb{R}^{(A-1)\times T\times 2}$ denote the predicted and ground-truth future positions, where A is the total number of agents including the ego agent.

Best-mode selection. During training, a single trajectory mode is selected by minimizing the displacement error in the position space:

$$m^* = \arg\min_{m} \sum_{t=1}^{T} \left\| \widehat{Y}_{m,t,1:2} - Y_{t,1:2} \right\|_2, \tag{1}$$

$$\widehat{\tau} = \widehat{Y}_{m^*}. \tag{2}$$

The selected mode index m^* is used both for ego-trajectory regression and for the classification target.

Ego imitation loss. The ego loss consists of a regression term and a classification term:

$$\mathscr{L}_{\text{reg}} = L1_{\text{smooth}}\big(\widehat{\tau}, Y\big), \tag{3}$$

$$\mathscr{L}_{\text{cls}} = \text{CE}\big(\pi, m^*\big). \tag{4}$$

Representing the heading as $(\cos\psi, \sin\psi)$ avoids angle wrap-around.

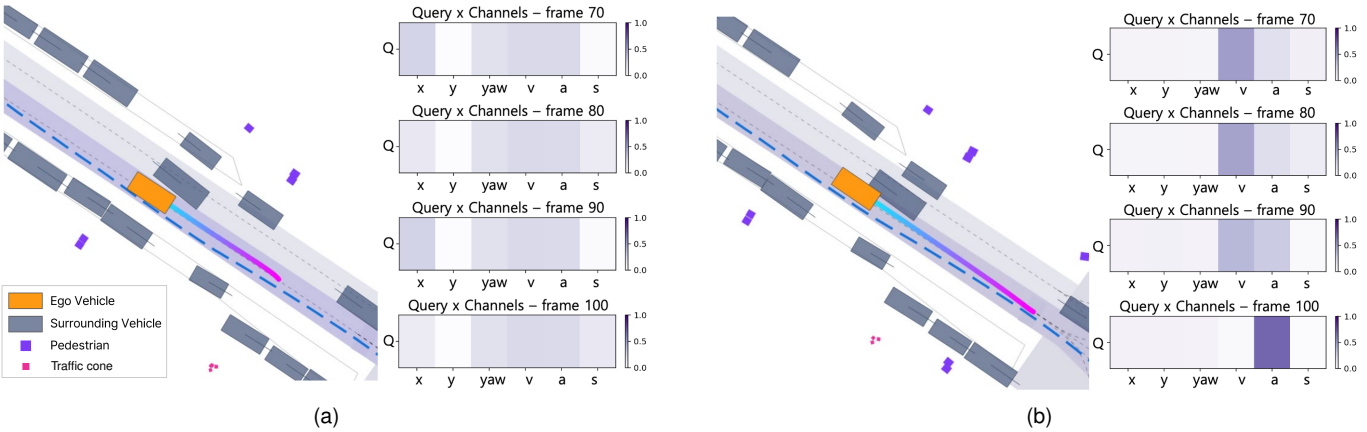


Fig. 2. **Ego-state attention comparison on nuPlan `test14-hard`, R-CLS.** (a) CAR. (b) PlanTF. The panels show the predicted trajectories and ego-state attention distributions for the two illustrated scenarios.

Non-ego prediction loss. Non-ego future positions are supervised with smooth-L_1 loss over valid agents and time steps:

$$\mathscr{L}_{\mathrm{p}} = L1_{\mathrm{smooth}}\big(\widehat{P}, P\big). \tag{5}$$

Task objective. The final task objective is

$$\mathscr{L}_{\mathrm{task}} = \mathscr{L}_{\mathrm{reg}} + \mathscr{L}_{\mathrm{cls}} + \mathscr{L}_{\mathrm{p}}. \tag{6}$$

This task loss is used in all three planners. CAR Planner further imposes the constraint described in the following subsection and handles it using ALM.

C. Problem formulation as a constrained optimization

Let $a_\theta^{(n)} \in \Delta^{C-1}$ denote the ego-state attention weights for the n-th sample, where a single learned query attends to C=6 ego-state channels. To discourage excessive concentration on only a few channels, the deviation of the attention distribution from uniform is measured as

$$d\big(a_\theta^{(n)}\big) = \frac{1}{C}\sum_{i=1}^{C}\Big|a_{\theta,i}^{(n)} - \tfrac{1}{C}\Big|. \tag{7}$$

Larger values indicate greater deviation from uniform attention. Averaging over a batch of N samples gives

$$D(\theta) = \frac{1}{N}\sum_{n=1}^{N} d\big(a_\theta^{(n)}\big). \tag{8}$$

The goal is to keep the batch-mean deviation $D(\theta)$ below a prescribed margin $\delta > 0$. A positive margin permits task-dependent, non-uniform attention without prescribing a fixed channel ranking. This leads to the inequality constraint

$$g(\theta) = D(\theta) - \delta \le 0, \tag{9}$$

where δ controls the allowed level of concentration. In the main experiment, δ=0.11 is used. A smaller value enforces more dispersed attention, while a larger value allows stronger selectivity. Frame-level dispersion quantiles and violation rates are evaluated separately in Section IV-B.

Figure 2 qualitatively illustrates the resulting behavior. Across the shown frames, CAR Planner retains non-uniform channel selectivity while exhibiting less concentration than PlanTF.

D. ALM-based update rule

The resulting optimization problem is

$$\min_\theta \ \mathscr{L}_{\mathrm{task}}(\theta) \quad \text{subject to} \quad g(\theta) \le 0, \tag{10}$$

which limits excessive ego-state attention concentration while preserving the original planning objective.

This constrained problem is optimized with the projected Powell–Hestenes–Rockafellar Augmented Lagrangian Method. Using a nonnegative Lagrange multiplier λ and penalty coefficient $\rho > 0$, the augmented objective is

$$\mathscr{L}_{\mathrm{aug}}(\theta,\lambda) = \mathscr{L}_{\mathrm{task}}(\theta) + \frac{1}{2\rho}\left([\lambda + \rho g(\theta)]_+^2 - \lambda^2\right). \tag{11}$$

Here, $[z]_+ := \max(0,z)$ denotes projection onto the nonnegative orthant. The effective constraint weight in the parameter update is $[\lambda + \rho g(\theta)]_+$. It depends on both the multiplier and the residual and can remain positive within the feasible region. Unless otherwise stated in the sensitivity study, $\rho = 3$ is used, and the multiplier is initialized as $\lambda_0 = 0$. At each iteration, the model parameters θ are updated by minimizing $\mathscr{L}_{\mathrm{aug}}$. After the optimizer step, the multiplier is updated using the same forward-pass residual:

$$\lambda_{t+1} = [\lambda_t + \rho g(\theta_t)]_+. \tag{12}$$

The constraint is applied only to the ego-state attention module, and the rest of the planner remains unchanged. At inference time, neither the penalty term nor the multiplier update is used, so the runtime architecture is identical to the base planner.

IV. EXPERIMENTS

A. Experimental setup

All experiments were conducted on a workstation equipped with an AMD EPYC 9354 32-Core Processor, 251 GiB of system memory, and four NVIDIA RTX PRO 5000 GPUs. Training uses a total batch size of 32 and Adam (learning rate 1e−3, weight decay 1e−4). The primary 1M CAR Planner runs use 25 epochs, while the controlled 100K regularizer comparison uses 20 epochs. During training, a state-perturbation augmentation [31] adds bounded noise to the

TABLE I
PERFORMANCE COMPARISON ON THE TEST14 BENCHMARK.

Planners		Test14-random			Test14-hard		
Type	**Method**	OLS	NR-CLS	R-CLS	OLS	NR-CLS	R-CLS
Expert	Log-replay	100.0	94.03	75.86	100.0	85.96	68.80
Rule-based	IDM	34.15	70.39	72.42	20.07	56.15	62.26
	PDM-Closed	46.33	90.05	91.64	26.43	65.08	75.19
Hybrid	PDM-Hybrid	81.26	90.10	91.29	74.08	65.99	76.07
Learning-based	RasterModel	62.93	68.71	67.52	52.38	50.65	51.41
	UrbanDriver	82.45	63.57	60.92	76.90	51.67	49.06
	GC-PGP	80.89	61.57	54.88	75.40	47.16	41.39
	PDM-Open	84.14	52.81	57.23	79.06	33.51	35.83
	Base	86.64	80.01	74.48	82.48	65.30	53.11
	PlanTF	86.27	85.23	79.36	83.34	70.03	59.83
	Pluto (w/o post.)	87.15	86.60	77.96	81.89	70.28	56.70
	CAR Planner (Ours)	87.64	85.25	79.15	86.09	72.94	64.90

current ego state, and all coordinates are re-normalized with respect to the perturbed ego frame.

Evaluation follows the nuPlan benchmark using two evaluation splits proposed by PlanTF and commonly adopted in nuPlan-based planning studies [10], `test14-random` and `test14-hard`. After applying the evaluation filters, `test14-random` contains 261 valid scenarios, whereas `test14-hard` contains 272 fixed scenario tokens selected from low-performing PDM-Closed candidates across 14 scenario types [14]. Reported metrics are OLS, NR-CLS, and R-CLS, where higher is better.

Parameter sensitivity is examined under a reduced 50K training budget. The penalty coefficients $\rho \in \{1,3,10\}$ are compared at $\delta = 0.11$ using three training runs per setting. The margins $\delta \in \{0.10, 0.11, 0.12\}$ are compared at $\rho = 3$ over five seeds on both test14 splits. The default setting is $\delta = 0.11$ and $\rho = 3$. Test14 performance was not used for checkpoint selection or early stopping, and all models were evaluated using their final-epoch checkpoints.

The benchmark includes the three variants defined in Section III-A and additional rule-based, hybrid, and learning-based planners. Table I groups Pluto without post-processing with the learning-based planners.

All reported baseline scores were obtained by evaluating the public implementations in the authors' environment. The learned baselines were also trained in that environment. Table II complements the test14 comparison with scenario-type R-CLS scores on val14.

The overall comparison uses one designated representative checkpoint per learned planner. Multi-seed auxiliary analyses are reported separately: ALM training dynamics for the primary 1M runs and score means and sample standard deviations for the 100K regularizer comparison.

B. Experimental results

Overall benchmark performance. Table I summarizes overall OLS, NR-CLS, and R-CLS performance on both `test14-random` and `test14-hard`. Among the evaluated learning-based planners, CAR Planner achieves the highest OLS on both splits and the highest NR-CLS and R-CLS on `test14-hard`. On `test14-random`, Pluto (w/o post.) and PlanTF obtain higher NR-CLS and R-CLS, respectively. The rule-based PDM-Closed and hybrid PDM-Hybrid obtain higher R-CLS on both splits.

TABLE II
R-CLS SCORES ON SIX SELECTED VAL14 SCENARIO TYPES. *N*: SCENARIO COUNT. BEST SCORES IN EACH ROW ARE SHOWN IN BOLD.

Scenario type	*N*	Base	PlanTF	CAR (Ours)
near_multiple_vehicles	85	70.58	75.83	**75.97**
waiting_for_pedestrian_to_cross	53	69.75	**74.19**	72.92
changing_lane	70	84.49	85.64	**86.91**
Intersection_traversal	98	73.07	66.04	**74.40**
stationary_in_traffic	100	83.89	88.83	**93.45**
high_magnitude_speed	100	78.48	**86.10**	82.47

Table II compares six val14 scenario types involving traffic, pedestrians, and varied maneuvers. CAR Planner exceeds Base in all six and performs comparably to or better than PlanTF in lane changes, intersection traversal, stationary traffic, and multi-vehicle settings. PlanTF leads in pedestrian waiting and high-speed driving. Together with the `test14-hard` results in Table I, these observations support robustness in challenging reactive closed-loop conditions.

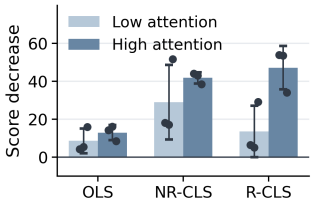
Constraint behavior and attention analysis. Table III compares attention concentration in representative Base and CAR Planner checkpoints trained on the full 1M-scenario set. In NR-CLS and R-CLS over 272 `test14-hard` scenarios, success and failure denote positive and zero final nuPlan scores, respectively, with groups formed separately for each model. Quantiles and violation rates pool recorded planning frames within each group, while counts refer to scenarios. A violation is defined by $d(a) > \delta = 0.11$. CAR Planner shows lower median dispersion and violation rates in every group. Although upper quantiles and maxima can still exceed the margin, the lower medians and violation rates indicate reduced attention concentration overall.

Velocity interventions on Base examine the association between attention concentration and channel-dependent vulnerability. Velocity receives the largest clean attention share,

TABLE III

FRAME-LEVEL ATTENTION CONCENTRATION OF BASE AND CAR PLANNER TRAINED ON THE FULL 1M-SCENARIO SET. STATISTICS POOL FRAMES WITHIN EACH MODEL'S SUCCESS AND FAILURE GROUPS OVER 272 `test14-hard` SCENARIOS.

Evaluation				Frame-level dispersion				
Model	**Setting**	**Outcome**	**Scenarios**	**Median**	p_{90}	p_{95}	**Max**	**Violation (%)**
Base	NR-CLS	Success	203	0.1535	0.1600	0.1601	0.1655	99.6
	NR-CLS	Failure	69	0.1532	0.1600	0.1601	0.1614	99.7
	R-CLS	Success	179	0.1532	0.1599	0.1601	0.1655	99.5
	R-CLS	Failure	93	0.1567	0.1600	0.1601	0.1608	98.9
CAR (Ours)	NR-CLS	Success	229	0.0978	0.1152	0.1515	0.1621	14.5
	NR-CLS	Failure	43	0.0977	0.1151	0.1516	0.1621	14.5
	R-CLS	Success	201	0.0978	0.1213	0.1545	0.1621	15.8
	R-CLS	Failure	71	0.0977	0.1112	0.1425	0.1621	11.0



(a) Velocity intervention

Comparison	NR-CLS	R-CLS
Attn.–Jac.	39.1	45.2
Attn.–Contrib.	43.2	49.2
Contrib.–Jac.	82.9	84.2

(b) Top-1 agreement (%)

Fig. 3. Base-only channel-dependence analysis. (a) Velocity-bias score decreases by clean-attention group across three 100K checkpoints on `test14-hard`. (b) Top-1 agreement for the representative full-1M Base checkpoint.

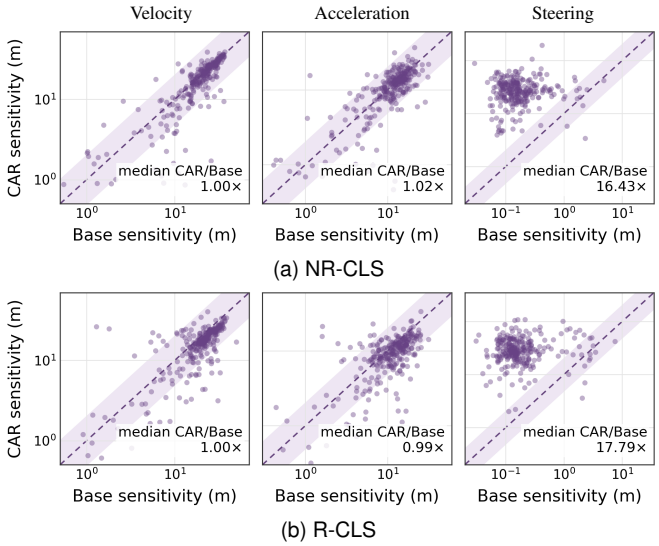


Fig. 4. Channel-wise trajectory response of full-1M Base and CAR Planner checkpoints on 272 paired `test14-hard` scenarios. Points represent scenario-mean scaled Jacobian norms, the diagonal denotes equal response, and shading marks CAR/Base ratios of 0.5–2.

approximately 0.50 across the closed-loop settings. Three fixed Base checkpoints trained on the 100K-scenario subset are evaluated with persistent $\pm 0.25\sigma_v$ biases, where $\sigma_v = 4.4891$ m/s is estimated from that training subset. Perturbed and original-input rollouts are paired by checkpoint and scenario over 272 `test14-hard` scenarios. The two bias signs are averaged within each scenario and checkpoint before aggregation.

In Fig. 3(a), the high-velocity-attention group shows a larger mean score decrease than the low-attention group in OLS, NR-CLS, and R-CLS. The groups are the upper and lower quartiles of clean attention within each checkpoint and evaluation setting. Bars and error bars report the mean and sample standard deviation across the three checkpoints. The difference is largest in R-CLS, with decreases of 13.6 and 47.2 for the low- and high-attention groups, respectively. Here, *shortcut-like channel reliance* denotes disproportionate degradation when a channel with dominant clean attention is biased.

The representative full-1M Base checkpoint is used to examine whether the most-attended channel also has the greatest output influence. Channel contribution is $c_i = \|W^O \mathrm{concat}_h(a_{h,i}V_{h,i})\|_2$, using per-head attention $a_{h,i}$, projected values $V_{h,i}$, and output projection W^O, excluding its bias. The scaled Jacobian is $j_i = s_i \|\partial \widehat{Y}_{xy}/\partial x_i\|_F$, where $\widehat{Y}_{xy}$ is the highest-probability mode's future position sequence, with its mode index held fixed. The norm spans future steps and both coordinates. For (x, y, ψ, v, a, s), the fixed scales are $(1\,\mathrm{m}, 1\,\mathrm{m}, 0.0873\,\mathrm{rad}, 1\,\mathrm{m/s}, 0.5\,\mathrm{m/s^2}, 0.05\,\mathrm{rad})$. These comparison scales are distinct from the empirical standard deviation used for input biases.

Top-1 agreement compares the highest-ranked of six channels at steps $0, 10, \ldots, 140$, then averages within scenarios and equally across scenarios. In Fig. 3(b), attention has 39.1–45.2% top-1 agreement with the input Jacobian, compared with 82.9–84.2% for value-aware contribution. This highlights the role of value representations and downstream processing in channel influence.

Figure 4 extends the analysis to the representative full-1M Base and CAR Planner checkpoints, comparing channel-wise trajectory responses through the input–trajectory Jacobian. Each point pairs scenario means of j_i over the sampled steps, expressed in meters, with velocity, acceleration, and steering shown from left to right. The diagonal denotes equal response, and the shaded band denotes a CAR/Base ratio of 0.5–2. Median ratios for velocity and acceleration remain close to one in both closed-loop settings. Steering responses are more pronounced in CAR Planner, with median ratios of 16.43 in NR-CLS and 17.79 in R-CLS. Thus, reduced concentration coexists with stronger steering responsiveness and similar median velocity and acceleration responses.

TABLE IV
CONTROLLED REGULARIZER COMPARISON UNDER MATCHED 100K TRAINING CONDITIONS. VALUES ARE MEAN SCORES AND SAMPLE STANDARD DEVIATIONS ACROSS SEEDS 0–9.

Regularization		Test14-random			Test14-hard		
Method	**Setting**	OLS	NR-CLS	R-CLS	OLS	NR-CLS	R-CLS
Base	–	83.82±0.60	69.04±3.55	68.58±2.92	80.38±0.40	54.97±2.94	51.41±1.65
Fixed hinge	β=0.2603	83.63±0.71	69.12±2.25	68.81±1.39	80.22±0.61	55.62±2.12	53.12±2.75
Fixed hinge (squared)	β=3.0388	83.24±0.37	69.45±2.24	69.92±2.21	80.28±0.46	55.30±2.27	53.39±2.77
Direct L1	β=0.2603	83.74±0.38	67.77±2.62	67.93±2.34	80.54±0.82	53.90±2.74	52.15±3.29
Direct L2	β=0.6230	83.31±0.47	67.98±1.39	67.81±2.06	80.65±0.66	53.36±2.45	51.43±1.90
Entropy/KL	β=0.03456	83.71±0.60	69.50±2.36	69.13±1.97	80.28±0.60	55.64±2.15	54.13±1.86
CAR (Ours)	δ=0.11, ρ=3	**84.61±1.14**	**69.90±2.03**	**70.75±2.20**	**81.98±2.00**	**57.38±2.19**	**55.36±1.96**

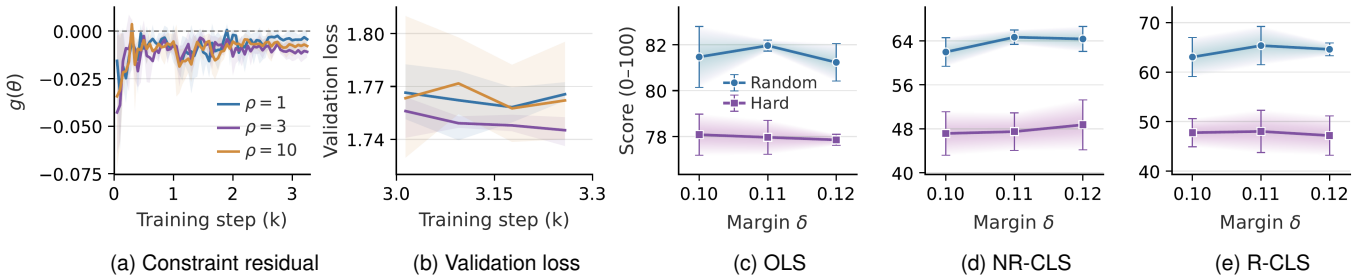


Fig. 5. Sensitivity to penalty coefficient and dispersion margin under 50K training. Panels (a)–(b) report means and min–max ranges over three runs, and panels (c)–(e) report mean scores and sample standard deviations over five seeds.

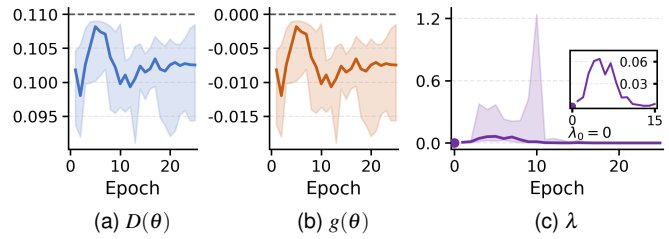


Fig. 6. ALM training dynamics of CAR Planner on the full 1M-scenario training set across seeds 0–9. Curves and bands show epoch-wise medians and P10–P90 ranges of dispersion, constraint residual, and multiplier.

These analyses distinguish attention allocation from output influence. CAR Planner regulates excessive allocation concentration through already-computed attention weights, providing a lightweight training signal without prescribing the contribution distribution. Contribution-based constraints involve value representations and output projection, while Jacobian regularization requires additional channel-wise gradients.

The constraint settings are compared in Fig. 5 under 50K training. Panels (a)–(b) compare ρ through the mean and min–max range of three runs. The mean residual settles below zero for all three settings, while $\rho = 3$ has a lower mean validation task loss and smaller run-to-run variation over steps 3,000–3,300. Panels (c)–(e) report margin-dependent scores as means and sample standard deviations over five seeds. The margin $\delta = 0.11$ gives the highest mean Random score in all three metrics, while Hard results vary with the metric. Considering both splits together with the validation-loss behavior motivates the default $\delta = 0.11$ and $\rho = 3$. These settings are empirical choices for the evaluated data and training budget and may require adjustment under other conditions.

These settings are then examined in the primary training runs. Figure 6 summarizes the epoch-wise median and P10–P90 range of batch-mean dispersion $D(\theta)$, signed residual $g(\theta) = D(\theta) - \delta$, and multiplier λ across seeds 0–9 trained on the full 1M-scenario set. Median dispersion stays below $\delta = 0.11$, with a negative median residual. The multiplier increases early in training and then approaches zero, while the percentile range shows variation across seeds. These trajectories illustrate the projected ALM update's response to the constraint residual.

Controlled regularization. Following the ALM analysis, Base, fixed hinge, squared hinge, Direct L1, Direct L2, Entropy/KL, and CAR Planner are compared using the same backbone, 100K training subset, optimizer, learning-rate schedule, batch size, and 20-epoch budget. Final-epoch checkpoints from seeds 0–9 are evaluated on identical `test14-random` and `test14-hard` scenarios. This study complements the primary 1M evaluation by comparing the full ALM construction with fixed-coefficient alternatives. PlanTF's SDE input dropout remains a full-scale reference, while this comparison focuses on direct attention regularization.

The fixed penalties are $\beta[D-\delta]_+$ (Fixed hinge), $\beta[D-\delta]_+^2$ (Squared hinge), βD (Direct L1), $\beta\langle(a_i - 1/C)^2\rangle$ (Direct L2), and $\beta\langle\sum_i a_i \log a_i\rangle$ (Entropy/KL). Averages span samples and, for L2, channels. Negative entropy equals $\mathrm{KL}(a\|u) - \log C$ for uniform u. Both hinge penalties share CAR Planner's dispersion and margin, with fixed coefficients and no multiplier update.

Fixed coefficients are calibrated to the initial CAR penalty's reference gradient ratio $r_{\mathrm{CAR}} = 0.0236214$, measured with $\lambda = 0$, $\delta = 0.11$, and $\rho = 3$. Each calibration uses a frozen Base checkpoint and 32 validation batches without

TABLE V
SCENARIO-LEVEL WALL-CLOCK RUNTIME ON 261 TEST14-RANDOM SCENARIOS. MEDIAN, MEAN, AND STANDARD DEVIATION SUMMARIZE VARIATION ACROSS SCENARIOS, WHILE TOTAL DENOTES ACCUMULATED RUNTIME.

Metric	CAR (Ours)	PlanTF	Urban	Raster	PDM-O	GC-PGP	PDM-H	PDM-C	Pluto (w/o post.)
Median (s)	14.97	15.15	17.09	18.07	19.05	21.76	32.65	34.32	45.39
Mean (s)	15.00	15.01	18.63	18.05	19.31	21.64	32.74	34.52	43.48
Std. (s)	2.45	2.43	6.27	6.31	6.86	6.89	8.44	9.09	13.89
Total (min)	65.2	65.3	81.1	78.5	84.0	94.1	142.4	150.2	189.1

Urban, Raster, PDM-O, PDM-H, and PDM-C denote UrbanDriver, RasterModel, PDM-Open, PDM-Hybrid, and PDM-Closed, respectively. Pluto is evaluated without post-processing.

parameter updates. For regularizer j, the coefficient is $\beta_j = r_{\text{CAR}}/\,\text{median}_b\, r_{j,b}(1)$, where $r_{j,b}(1)$ is the regularizer-to-task gradient L2 norm ratio over ego-state encoder parameters at $\beta = 1$. The separate squared-hinge calibration retains this reference and yields $\beta = 3.038757$. Coefficients remain fixed during training.

Table IV reports means and sample standard deviations across ten seeds. Squared hinge improves mean R-CLS over linear hinge on both splits, while CAR Planner retains the highest mean in all six settings. In Hard R-CLS, CAR Planner also has lower seed-to-seed variation than either hinge penalty. Unlike fixed-coefficient alternatives, ALM updates its multiplier using the signed constraint residual. The results support the benefit of the full ALM construction beyond the tested linear and quadratic threshold penalties.

Inference efficiency. Table V reports cumulative wall-clock runtime per scenario on 261 `test14-random` scenarios under the same hardware and software environment. This measure covers repeated planner calls and simulation and framework overhead. CAR Planner averages 15.00 seconds per scenario (65.2 minutes in total), comparable to PlanTF's 15.01 seconds (65.3 minutes). The attention constraint and multiplier update operate only during training, leaving inference computation unchanged. Together with the preceding performance results, this comparison shows improved planning performance in the evaluated challenging settings with comparable scenario-level runtime.

Qualitative example. Figure 7 presents a selected closed-loop scenario from the nuPlan `test14-hard` split. PlanTF and CAR Planner are shown at matching frame indices, with driving scenes and ego-state attention heatmaps illustrating the evolution of vehicle motion and channel allocation.

V. CONCLUSION

This paper introduced CAR Planner, a constrained-attention-based framework designed to reduce excessive ego-state attention concentration while retaining task-dependent selectivity. ALM dynamically adjusts constraint pressure according to the constraint residual during training. The formulation operates without discarding input information or imposing uniform attention across channels.

On nuPlan, CAR Planner reduces attention concentration, while channel analyses associate concentrated attention with input-dependent vulnerability. It remains competitive on the random split and achieves the highest hard-split closed-loop scores among the evaluated learning-based planners. Controlled multi-seed comparisons further show higher mean scores than fixed-coefficient attention regularizers under matched reduced-data training conditions. Together, these findings support adaptive attention constraints as a lightweight approach with planning performance benefits under the evaluated conditions and no additional inference-time computation.

The current formulation is limited to ego-state cross-attention and has been validated only on nuPlan. Extending the constraint to interaction and map-level features, and validating on broader datasets or real-world testbeds, remains an important direction for future work.

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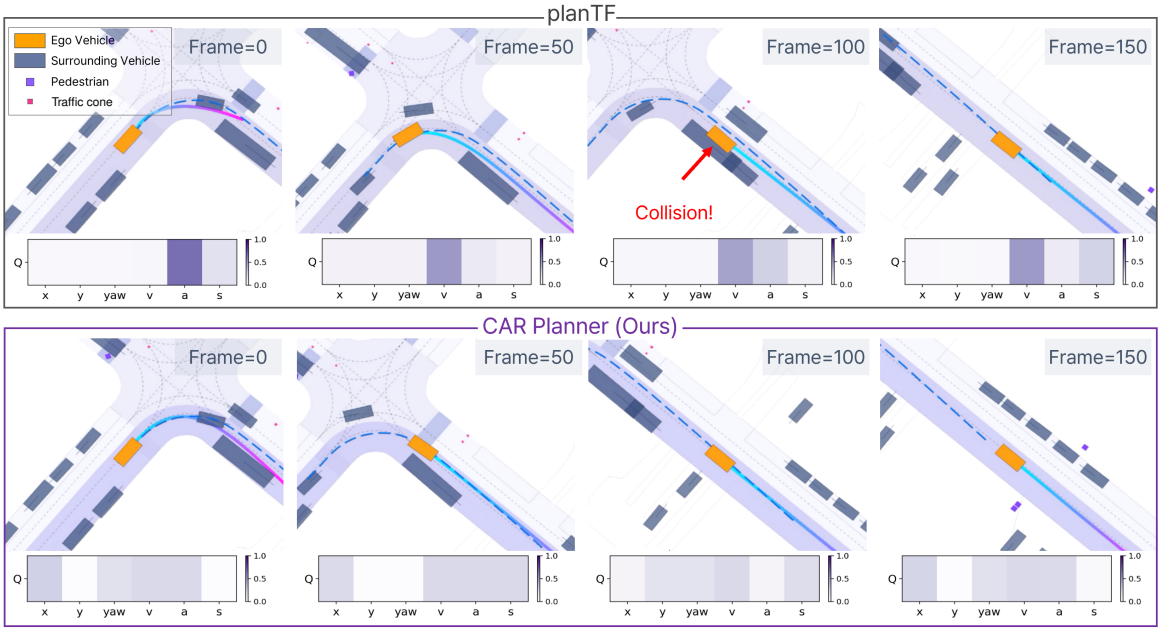


Fig. 7. Closed-loop simulation of a selected nuPlan `test14-hard` scenario. PlanTF (top) and CAR Planner (bottom) are shown at matching frame indices with driving scenes and ego-state attention heatmaps.

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